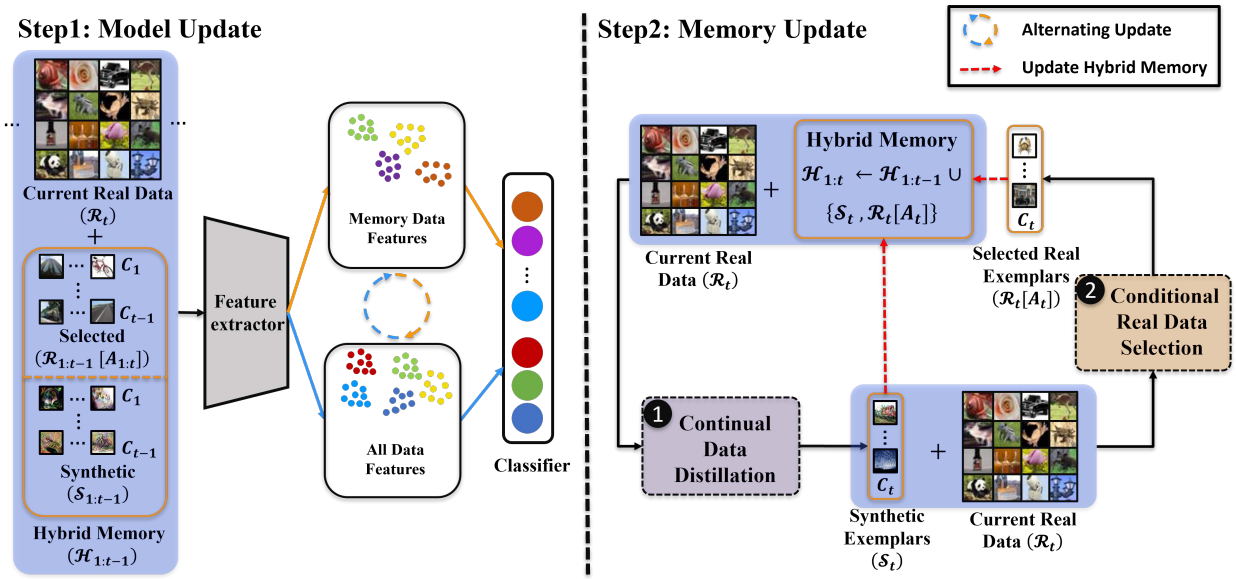

Figure 3: The framework of the proposed hybrid memory system for replay-based CIL. We first leverage the current real data with the hybrid memory for former classes to update the model. Then we use Continual Data Distillation (❶) to extract synthetic exemplars and conditional real data selection (❷) to choose optimal exemplars conditioned on synthetic data. Finally, the synthetic exemplars and selected real exemplars are combined to update the hybrid memory.

comprises synthetic exemplars for the classes in task-t, and $A_t \subseteq V_t$ ($V_t = \{1, 2, \cdots, |\mathcal{R}_t|\}$ is the ground set) indexes a subset of real samples selected from $\mathcal{R}_t$. The model $f(x;\theta)$ trained on $\mathcal{H}_{1:t-1} \cup \mathcal{R}_t$ is required to predict probabilities for all former classes, including the current task-t, denoted as $C_{1:t} \triangleq \bigcup_{i=1}^{t} C_i$, for any given input x. **We define the important notations used in this work in Appendix A**.

4.2 Overview of the Proposed Hybrid Memory

Fig. 3 presents the framework of the proposed hybrid memory system for CIL, comprising two key components: (i) model update and (ii) memory update. The model update refines the CIL model by leveraging a hybrid memory composed of both synthetic and real exemplars. Instead of trivially merging synthetic and real data, our memory update jointly optimizes the two by selecting real exemplars based on the optimized synthetic data. As the memory update is central to our method, we first elaborate on this component below.

4.3 Memory Update

According to Section 3, we propose to obtain an optimal hybrid memory $\mathcal{H}_t^\star$ for task-t by optimizing the following objective,

$$\mathcal{H}_t^\star \in \arg\min_{\mathcal{H}_t} \mathcal{L}(\mathcal{H}_t, \mathcal{R}_t), \quad \mathcal{H}_t \triangleq \{\mathcal{S}_t; \mathcal{R}_t[A_t]\}, \tag{1}$$

where $\mathcal{L}(\cdot,\cdot)$ measures the distance between the hybrid memory $\mathcal{H}_t$ and the real dataset $\mathcal{R}_t$. The hybrid memory minimizing $\mathcal{L}(\cdot,\cdot)$ is expected to achieve comparable performance as $\mathcal{R}_t$ when used to train the model.

However, jointly optimizing both the synthetic exemplars $\mathcal{S}_t$ and the subset A_t of real exemplars at the same time is challenging, as the optimization of $\mathcal{S}_t$ is a continuous process while selecting real exemplars $\mathcal{R}_t[A_t]$ requires combinatorial optimization. To address this problem, we adopt an alternating optimization strategy to optimize the hybrid memory, as illustrated on the right side of Fig. 3. Below, we detail our method for constructing an effective hybrid memory.

❶ **Continual Data Distillation (CDD).** First, we aim to optimize the synthetic exemplars $\mathcal{S}_t^\star$ for hybrid memory. Traditional DD methods can condense dataset information into a few synthetic exemplars, but they cannot be directly applied to replay-based CIL methods. This is because existing DD methods distill synthetic data from multiple training trajectories (with different random initialization) while most CIL methods only produce one trajectory per task. To address this issue, we introduce a novel Continual Data Distillation (CDD) technique, which distills $\mathcal{S}_t$ from checkpoints on task-t's training trajectory $\theta_{1:N}$, where N represents the total number of cached checkpoints θ in the

history of learning task-t. We do not need multiple trajectories because the replay of $\mathcal{S}_t$ in the next task will only start from θ_N instead of random initializations.

Specifically, to distill synthetic exemplars from the real dataset (❶) after training the model on task-t, we will minimize the distance ($\mathcal{L}(\cdot,\cdot)$ in Eq. 1) between synthetic exemplars and the real data from task-t. To effectively solve this equation, we propose CDD to adapt the original DD optimization objective into CIL below,

$$\mathcal{S}_t^\star \in \arg\min_{\mathcal{S}_t} \mathcal{L}(\mathcal{S}_t, \mathcal{R}_t) = \mathop{\mathbb{E}}_{\theta\sim\theta_{1:N}} \left[\ell(\mathcal{S}_t, \mathcal{R}_t; \theta)\right], \tag{2}$$

where $\ell(\cdot,\cdot;\cdot)$ in Eq. 2 is a DD objective from any existing DD methods. It aims to measure the distance between the synthetic exemplars $\mathcal{S}_t$ and the real data $\mathcal{R}_t$ using the sampled model parameters θ. **In Appendix D, we present a detailed description of several optimization objectives for CDD in CIL.**

To keep a constant and smaller memory of model checkpoints, we can instead apply a sliding window version of CDD of window size $\tau \ll N$, i.e., updating $\mathcal{S}_t$ only based on the most recent τ checkpoints. Specifically, after each epoch-$j+1$ of task-t, we update the synthetic data $\mathcal{S}_t$ as $\mathcal{S}_{t,j+1}$ below.

$$\mathcal{S}_{t,j+1} = \mathcal{S}_{t,j} - \eta \nabla_{\mathcal{S}_{t,j}} \left(\mathop{\mathbb{E}}_{\theta\sim\theta_{j-\tau+2:j+1}} \left[\ell(\mathcal{S}_{t,j}, \mathcal{R}_t; \theta)\right] \right), \tag{3}$$

where η is the learning rate for updating the synthetic exemplars $\mathcal{S}_{t,j+1}$. Hence, we only need to store τ checkpoints in any step but the final $\mathcal{S}_t$ at the end of each task is a result of DD on all the N checkpoints. The final $\mathcal{S}_t^\star \leftarrow \mathcal{S}_{t,N}$.

❷ **Conditional Real Data Selection**. While synthetic exemplars effectively capture rich information from previous tasks, their impact tends to plateau as their quantity increases (see Sec. 3). To overcome this limitation, we select a subset of real data complementary to $\mathcal{S}_t^*$ at the end of each task t to further reduce the gap between the memory $\mathcal{H}_t$ and $\mathcal{R}_t$ (refer to ❷ in Fig. 3). Guided by Eq. 1, the objective is to find an optimal subset $A_t^\star \subseteq V_t$ such that the selected real data $\mathcal{R}_t[A_t^\star]$, in conjunction with the distilled synthetic exemplars $\mathcal{S}_t^\star$, minimize the DD objective below:

$$A_t^\star \in \mathop{\arg\min}_{A_t\subseteq V_t, |A_t|\leq k} \ell([\mathcal{R}_t[A_t]; \mathcal{S}_t^\star], \mathcal{R}_t; \theta_t), \tag{4}$$

where θ_t denotes the model parameters trained on $\mathcal{R}_t \cup \mathcal{H}_{1:t-1}$, task-$t$'s real data $\mathcal{R}_t$ combined with the hybrid memory $\mathcal{H}_{1:t-1}$ for the former classes. The DD optimization objective $\ell(\cdot,\cdot;\cdot)$ is kept consistent with the one in Eq. 2.

Since the conditional subset selection problem in Eq. 4 is an NP-hard combinatorial optimization problem, we apply the greedy algorithm to approximately optimize the subset A_t.

Algorithm. To select some ideal real exemplars, we employ a greedy algorithm within our selection. This approach involves iteratively selecting the locally optimal real exemplars from the current task, guided by the objective in Eq. 4. The core idea is to approximate a globally optimal selection, designated as $\mathcal{R}_t[A_t^\star]$, through these successive local optimizations. **The detailed algorithm is summarized in Appendix B.**

Theoretical Analysis. We also theoretically analyze the performance of the model trained on the proposed hybrid memory, showing that it can achieve performance comparable to a model trained on the real dataset for all tasks in CIL. We begin our analysis by learning CIL tasks jointly, which serves as the performance benchmark for CIL. In this scenario, the training data for task-t is written as $\mathcal{R}_{1:t} \triangleq \bigcup_{i=1}^{t-1} \mathcal{R}_i \cup \mathcal{R}_t$ and the model for task-t can be obtained by the maximum likelihood estimation below

$$\theta_{\mathcal{R}_{1:t}} = \arg\max_\theta (\log P(\mathcal{R}_{1:t-1}|\theta) + \log P(\mathcal{R}_t|\theta)). \tag{5}$$

Given that the absence of $\mathcal{R}_{1:t-1}$ leads to catastrophic forgetting in CIL, our objective is to construct a smaller hybrid memory, $\mathcal{H}_{1:t-1}$, with limited exemplars that enable the network to achieve performance comparable to that obtained when trained on $\mathcal{R}_{1:t-1}$.

Definition 1. *The optimal model trained on task-t's hybrid memory is* $\theta_{\mathcal{H}_t} = \arg\max_\theta \log P(\mathcal{H}_t|\theta)$.

Definition 2. *The optimal model trained on task-t's real data is* $\theta_{\mathcal{R}_t} = \arg\max_\theta \log P(\mathcal{R}_t|\theta)$.

Definition 3. *The optimal model trained on the hybrid memory of all prior tasks from 1 to t is denoted by* $\theta_{\mathcal{H}_{1:t}} = \arg\max_\theta h_t(\theta) = \arg\max_\theta \log P(\mathcal{H}_{1:t}|\theta)$.

Definition 4. *The optimal model trained on the real data of all tasks from 1 to t is defined as* $\theta_{\mathcal{R}_{1:t}} = \arg\max_\theta r_t(\theta) = \arg\max_\theta \log P(\mathcal{R}_{1:t}|\theta)$.

Assumption 1. *Based on our objective of the proposed hybrid memory, we assume that the model trained on the hybrid memory for one task can achieve comparable performance to that of the model trained solely on real data for the same task, we can express this formally as* $\exists \epsilon_t \in [0,1), \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{H}_{t+1}}) \geq (1-\epsilon_t)\log P(\mathcal{R}_{t+1}|\theta_{\mathcal{R}_{t+1}})$.

Assumption 2. *Assume that the performance of the model trained on the hybrid memory of all prior tasks till task-t and the performance of the optimal model trained on the hybrid memory of task-*$t+1$ *can be bonded as* $\exists \rho, \rho * r_{t+1}(\theta_{\mathcal{R}_{1:t+1}}) \geq r_t(\theta_{\mathcal{H}_{1:t}}) + \log P(\mathcal{R}_{t+1}|\theta_{\mathcal{H}_{t+1}})$.

Theorem 1 (Performance Approximation)**.** Based on the above Assumptions 1 and 2, when $\epsilon_{t+1} \geq \frac{\rho}{1-\epsilon_t}$, we can derive that the model trained on the hybrid memory of all previous tasks achieves performance comparable to that of the model trained on the real dataset of all previous tasks,

$$\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{H}_{1:t}}) \geq (1-\epsilon_t)\log P(\mathcal{R}_{1:t}|\theta_{\mathcal{R}_{1:t}}). \tag{6}$$

We provide the detailed proof of Theorem. 1 in Appendix. C.

Remark. *According to Theorem 1, when* ϵ_t *and* ρ *are small, Eq.6 is well bounded. In other words, if the proposed hybrid memory is optimized using* $\mathcal{L}(\cdot,\cdot)$ *in Eq.1 for each task in CIL, the performance of the model trained on the hybrid memory across all tasks can achieve results comparable to those of a model trained on a real dataset for all tasks.*

4.4 Model Update

Next, we describe the model update based on the updated hybrid memory. After optimizing the hybrid memory $\mathcal{H}_t$ for the task-t, the hybrid memory is then updated as follows:

$$\mathcal{H}_{1:t} \leftarrow \mathcal{H}_{1:t-1} \cup \{\mathcal{S}_t^\star, \mathcal{R}_t[A_t^\star]\}. \tag{7}$$

We then combine the updated hybrid memory, $\mathcal{R}_{t+1}$, which contains data from previous tasks, with the real exemplars, $\mathcal{R}_{t+1}$, to train the CIL model for task-$t+1$. As illustrated on the left side of Fig.3, to achieve the optimal model trained on the hybrid memory of all prior tasks, i.e., $\theta^H_{1:t}$ in Eq. 6, we adopt an alternating model update strategy based on the data and hybrid memory of task-t.

Finally, we detail the training pipeline for task-t in Algorithm 1.

Algorithm 1: Training with Hybrid Memory on Task-t

input : Epochs: N; sliding window: τ; real data for classes C_t: $\mathcal{R}_t$; number of synthetic exemplars per class: k; hybrid memory from previous tasks: $\mathcal{H}_{1:t-1}$

```
Randomly select k samples per class from R_t as S_{t,0};
for j ∈ {1, ..., N} do
    update the model as θ_j on R_t ∪ H_{1:t-1} using a specified CIL objective and cache θ_j;
    if j > τ then
        remove the cached checkpoint θ_{j-τ};
        update synthetic data S_{t,j} using Eq. 3;
    else
        S_{t,j} ← S_{t,j-1};
S_t ← S_{t,N};
optimize A_t in Eq. 4 using Algorithm 2 to obtain the selected real exemplars as R_t[A_t];
```

output : Hybrid Memory $\mathcal{H}_{1:t} \leftarrow \mathcal{H}_{1:t-1} \cup \{\mathcal{S}_t, \mathcal{R}_t[A_t]\}$

5 Experiments

In this section, we first compare the performance of the proposed hybrid memory with the replay-based baselines in CIL. We then demonstrate that the integration of the hybrid memory can enhance the performance of some existing replay-based CIL methods. Additionally, we investigate the ratio of synthetic exemplars within the hybrid memory. Finally, we examine the effectiveness of our conditional real data selection in choosing ideal real exemplars.

Datasets. We evaluate the proposed method on two commonly used benchmarks for class incremental learning (CIL), CIFAR-100 [14] and TinyImageNet [37]. ***CIFAR-100:*** CIFAR-100 contains 100 classes, the whole dataset includes